

IIP Kiosk Final Year Project Change Summary

Generated on 29 June 2026.

This report combines the previous requested changes with the latest public Pulse display and feedback leaf-flight animation updates.

The goal is to document what was changed, what bugs were fixed, and how the code supports the kiosk system.

Latest Changes in This Update

- Reworked the Pulse screen for a 7m by 2m style display so the tree, metrics, campaign progress, badge board, pledge feed, and badge mix are visible without page scrolling.
- Kept Pulse public across backend/server.js, backend/adminServer.js, and backend/kioskServer.js by removing Pulse auth middleware from the page and API routes.
- Updated admin wording from admin-only Pulse to public display Pulse so the interface matches the route behavior.
- Improved the feedback completion animation with real tree leaf images, badge-specific flight paths, motion trails, and a canopy glow when the leaf lands.
- Updated the PDF documentation to explain the code and system behavior for final year project review.

Previous Feedback Screen Changes

- Removed the Quick Consent page from the visitor feedback flow.
- All feedback submissions now save with long-term retention from both frontend and backend.
- Removed Data Retention and Primary Consent from the Confirm Submission page list.
- Added an admin feature switch called Thank You Social Buttons to hide or show social media buttons on the Thank You page.
- Merged the separate badge email and thank-you/photo email into one combined visitor email response.
- The combined thank-you email can still include or hide badge details using the badge email feature flag.

Visitor Passport Feature

- Added POST /api/feedback/visitor-passport in backend/feedbackRoutes.js.
- The endpoint maps a visitor passport to the submitted email address by decrypting stored encrypted emails and matching the current visitor email.
- The response includes masked email, holder name, visit count, feedback count, pledge count, photo keepsakes, badge collection, passport stamps, and recent visits.
- The Thank You page calls the passport endpoint after submission and renders collected badges and recent visit stamps.
- The current visit is shown immediately, then merged with backend history after database save. The frontend prevents duplicate current-visit entries.

Pulse Screen Public Ultrawide Layout

- frontend/pulse/pulse.css now uses height: 100vh and overflow: hidden so the public screen cannot scroll.
- For ultrawide screens, CSS grid areas place the tree on the left, metrics and bottom pulse panels in the middle, and campaign/badge panels on the right.
- The layout is designed for the physical 7m by 2m display ratio while still preserving the same data from /api/pulse/summary.
- frontend/pulse/pulse.js continues to fetch live data every 20 seconds and render metrics, pledges, badge earners, badge mix, and tree visitors.
- backend/pulseRoutes.js returns public dashboard data without exposing visitor emails. Names are shown only where long-term display consent is saved.

Feedback Leaf-Flight Animation

- frontend/feedback/feedback.html adds target and trail elements inside the Thank You celebration tree preview.
- frontend/feedback/feedback.css animates a real leaf image along a curved path into the mini tree using celebrationFlyLeaf, celebrationFlyLeafTrail, and celebrationTreeCatch keyframes.
- frontend/feedback/feedback.js maps each badge type to a leaf image and landing target, so Climate Champion, Ocean Guardian, Social Champion, and other badges fly into slightly different canopy positions.
- The animation starts when setupCelebrationMoment adds the is-ready class after a successful feedback submission.
- The result gives the visitor visual proof that their feedback has become part of the digital tree.

Tree Screen Changes

- Removed the feedback box below each Tree screen leaf.
- Added a leaf appearance animation when leaves are created.
- Kept the leaf dropping animation and connected timing to CSS variables so the behavior can be tuned safely.

Bugs Found and Fixed

- Fixed backend/feedbackRoutes.js existing-user email update bug where encryptedEmail could be referenced outside its scope.
- Fixed backend/kioskServer.js still requiring authentication for Pulse even after Pulse was meant to be public.
- Updated tests/pulse-backend-check.js so automated checks expect Pulse to be public.
- Fixed visitor passport refresh logic so the current visit is not duplicated when backend history already includes it.
- Fixed retry-email behavior so retried thank-you emails use the same merged badge include flag, badge key, and badge color as normal submissions.
- Removed stale active-path response/UI references to badgeEmailSent, badgeEmailError, and badgeEmailBadgeKeys.
- Confirmed axios is present in package.json, which resolves the earlier Cannot find module axios error after dependencies are installed.
- Checked port 3000 after testing; no listener remained. If EADDRINUSE returns later, it means another local process is already using port 3000.

How the Main Code Flow Works

- A visitor completes feedback in frontend/feedback/feedback.js. finalSubmit sends user data, answers, photo details, theme, and long-term retention to /api/feedback/submit-feedback.
- backend/feedbackRoutes.js validates the request, immediately returns a success response for a fast kiosk experience, then saves user and feedback records in the background.
- After the database save, the backend queues the combined thank-you/photo/badge email if email and photo data are available.
- The Thank You page calls setupCelebrationMoment for the badge and tree animation, setupJourneyPassport for visit stamps, and setupSocialShare only if the admin social toggle is enabled.
- Pulse uses frontend/pulse/pulse.js to call /api/pulse/summary. backend/pulseRoutes.js aggregates counts and visitor display data from feedback and users tables.
- Tree display uses frontend/tree/tree.js and backend/treeRoutes.js to render active, non-archived visitor leaves.

Verification Run

- node --check passed for the changed frontend and backend JavaScript files.
- Full non-node_modules JavaScript syntax sweep previously passed for 34 files.
- npm.cmd run test:pulse passed against a temporary backend/server.js process.
- The Pulse smoke test confirmed backend status, database connection, and public /api/pulse/summary payload.
- Confirmed the temporary server was stopped afterward and no node.exe process remained running.
- git diff --check previously passed with line-ending warnings only.

Files Changed

- backend/feedbackRoutes.js
- backend/emailService.js
- backend/server.js
- backend/adminServer.js
- backend/kioskServer.js
- backend/pulseRoutes.js
- frontend/feedback/feedback.html
- frontend/feedback/feedback.css
- frontend/feedback/feedback.js
- frontend/pulse/pulse.html
- frontend/pulse/pulse.css
- frontend/pulse/pulse.js
- frontend/admin/admin.html
- frontend/admin/admin.js
- frontend/tree/tree.js
- frontend/tree/tree.css
- tests/pulse-backend-check.js
- docs/change-summary-2026-06-29.pdf

FYP Notes

- The system separates visitor-facing routes from admin-only routes. Pulse is public because it is meant for a public display screen.
- Personal visitor passport lookup uses the submitted email as the key, but the stored database email remains encrypted and the returned email is masked.
- The Pulse display is read-only and does not provide admin actions, so public access is limited to dashboard data.
- The animation and passport features improve user feedback by showing immediate impact after form submission.